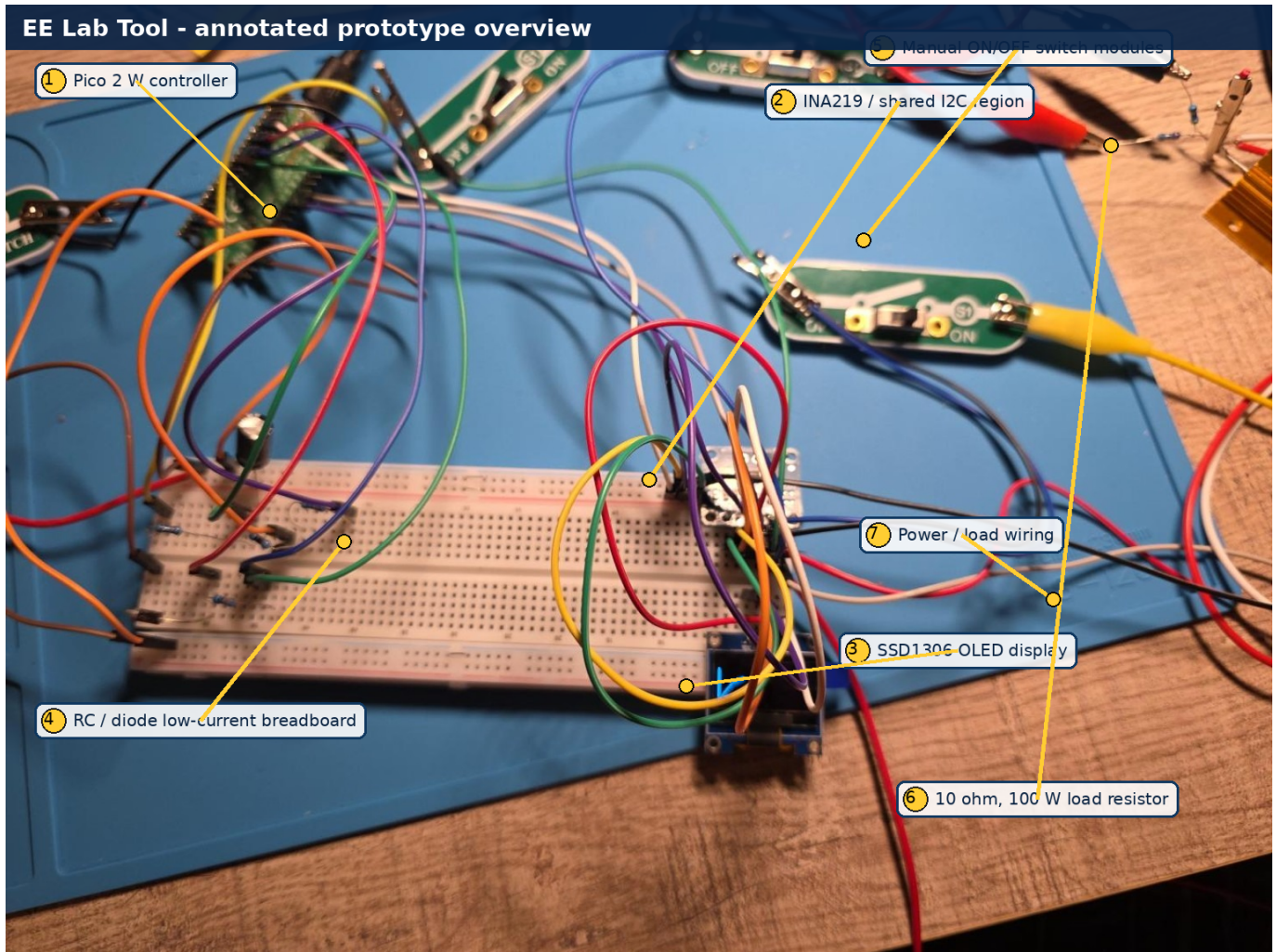


EE Lab Tool - Annotated, 2D, and 3D Design Assets

The verified 2D architecture diagram is authoritative for pin assignments. Conceptual renders are illustrative and are labeled accordingly.

Annotated prototype overview

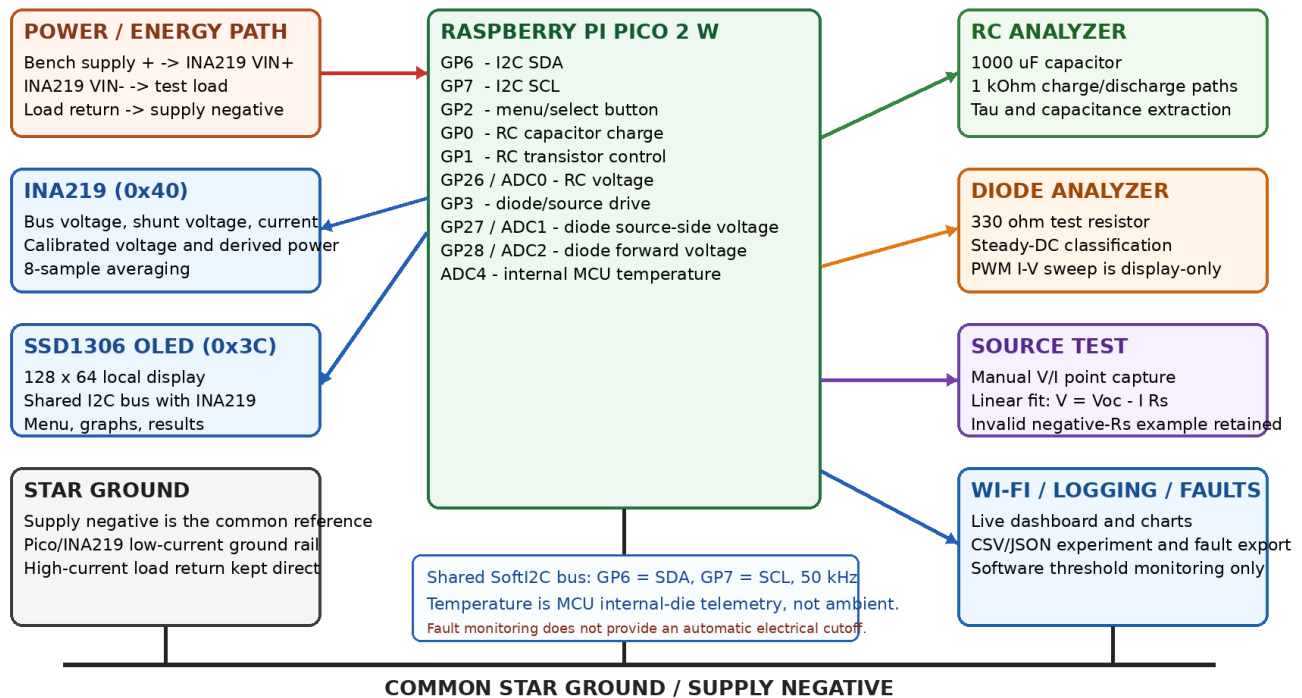


Real sanitized prototype photograph with subsystem callouts.

Verified 2D system architecture

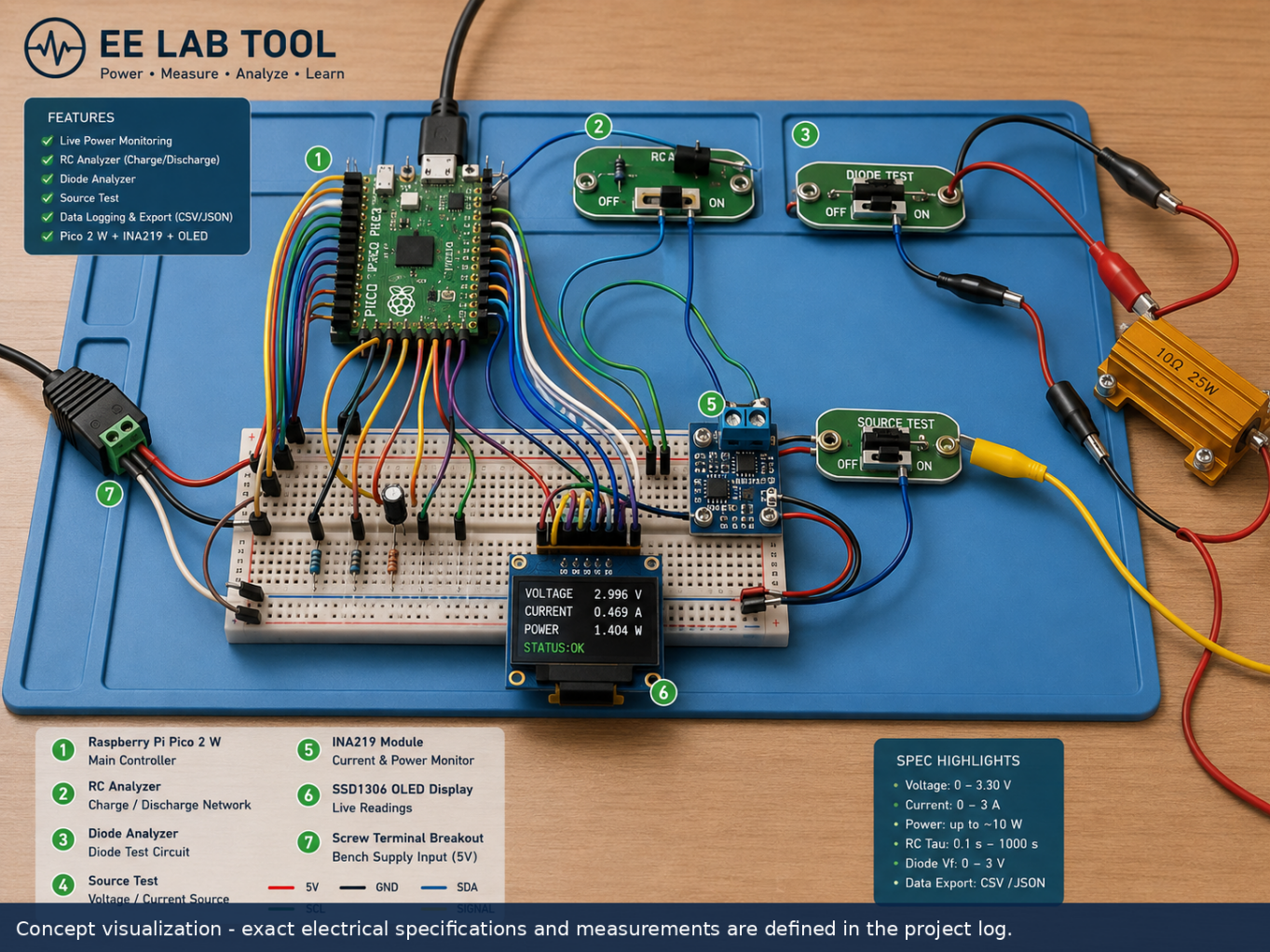
EE LAB TOOL - VERIFIED SYSTEM ARCHITECTURE

Raspberry Pi Pico 2 W | MicroPython | SoftI2C 50 kHz | Wi-Fi dashboard



Exact pins, SoftI2C frequency, measurement blocks, and system boundaries.

Conceptual 3D visualization



Presentation-oriented isometric design. Exact specifications are in the project log.

Power/Energy Monitor • RC/Diode Analyzer • MCU Telemetry • Source Test • Fault Monitoring • Experiment Logging

VOLTAJE

2.996 V

CURRENT

0.469 A

POWER

1.404 W

Power Monitor

CHARGE	ENERGY
9.70 mAh	0.0292 Wh
ELAPSED	MEDIAN VOLTAGE
103 s	2.992 V
MAXIMUM VOLTAGE	MEAN CURRENT
3.301 V	0.469 A
PEAK POWER	AVERAGE VOLTAGE
1.405 W	3.064 V
AVERAGE CURRENT	AVERAGE POWER
0.361 A	1.086 W

RC Analyzer + Diode Analyzer

RC Analyzer

VOC

-0.003 V

TAU

1.079 s

CAPACITANCE

1079 μ F

LAST RUN

278 s ago

Diode Analyzer

CLASSIFICATION

SILICON

FORWARD VOLTAGE

0.662 V

TEST CURRENT / SOURCE

7.09 mA / 3.000 V

LAST RUN

4 s ago

Source Test

VOC

-0.003 V

SOURCE RESISTANCE

-6.395 ohm

INVALID / REQUIRES REFINEMENT

FIT / POINTS

1.000 / 3 pts

Recent Power History

Voltage (V)

Time since boot (s)

119

Current (A)

Time since boot (s)

119

Power (W)

Time since boot (s)

119

RC/MCU Telemetry

Charge Voltage (V)

Elapsed time (s)

5.97

Discharge Voltage (V)

Elapsed time (s)

5.97

MCU Temperature (°C)

Time since boot (s)

307

(Internal die temperature)

Experiment History / Logs

Download Experiment CSV

Download Experiment JSON

Download Fault CSV

Download Fault JSON

Type	Status
RC	completed
POWER	completed
SOURCE TEST	completed
DIODE_SINGLE	completed
SOURCE TEST	completed
DIODE_IV	display_only
RC	completed
MC	completed
POWER	completed
FAULT	active

Updated 11:10:51 PM

Presentation asset: use the text documents as the authoritative source.

Conceptual project timeline visual

